

[69] SEAT No. _____

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SARDAR PATEL UNIVERSITY

External Examination 2021

B.sc.CA & IT Semester -III

PS03CIIT27: OPERATING SYSTEMS

Date: 26.11.2021 Time: 12:00 P.M to 2:00 P.M Total Mark: 70

Q1 Answer following MCQ. 10

1. Compiler is an example of which components of Operating system?
(a) Hardware (b) System Program (c) Application Program (d) None
2. Patient Monitoring in hospital is an example of which type of OS?
(a) Multi User (b) Real Time (c) Time Sharing (d) None
3. _____ types of OS used in scientific research and robotics
(a) Real-time (b) Time-sharing (c) Multi-user (d) None of these
4. In Paging, Physical memory is divided into fixed-size blocks is called _____.
(a) Files (b) Pages (c) Frames (d) None of these
5. Round Robin scheduling is essentially the preemptive version of _____.
(a) SJF (b) FCFS (c) Priority (d) None of these
6. _____ page replacement algorithm suffers from Belady's anomaly.
(a) FIFO (b) Optimal (c) LRU (d) None of these
7. In Clock page replacement algorithm, replace the page with _____.
(a) R-bit = 1 (b) R-bit = 0 (c) Valid bit = 0 (d) None of these
8. Which of the following memory allocation strategy is fastest?
(a) Best-fit (b) Worst-fit (c) Optimal-fit (d) First-fit
9. _____ command is use to display help of LINUX commands.
(a) helptop (b) mv (c) rm (d) man
10. _____ option of date command will display full month name.
(a) -a (b) -b (c) -B (d) None

Q2 Answer following Question . 08

1. The If statement ends with if. [True/False]
2. Printer is an example of _____ resource.
3. Preemption is not a valid deadlock characteristic. [True/False]
4. Demand Paging is Swapping with fragmentation. [True/False]
5. Valid-Invalid bit scheme in page table is used in _____
6. Short-term Scheduler select process from ready queue for execution. [True/False]
7. Time-sharing system also known as _____.
8. _____ scheduling algorithm gives minimal average waiting time.

(1)

P.T.O.

- Q3 Answer following Questions.(Attempt Any 10) 20**
- 1 Define: Operating system, Throughput, Turnaround Time, Response Time, Waiting time
 - 2 Explain Key elements of OS.
 - 3 Explain any two process scheduling criteria in brief.
 - 4 Explain Internal Fragmentation in short.
 - 5 What is Belady's Anomaly?
 - 6 What is Compaction? For what purpose it will use?
 - 7 Explain any two necessary conditions of Deadlock.
 - 8 Why is LINUX an open source?
 - 9 Explain resource utilization in details.
 - 10 Explain who command in brief.
 - 11 Explain ls -l command.
 - 12 Explain Test command in short.
- Q4 Explain in detail the following Questions.(Attempt any 4) 32**
- 1 Which are the functions performed by Operating System?
 - 2 Explain Priority Scheduling algorithm in brief
 - 3 What is Memory Management? Explain Logical-Versus-Physical Address space.
 - 4 Define Page Fault. Explain page fault handling technique.
 - 5 What is Process Synchronization? Explain
 - 6 What are the necessary conditions for Deadlock prevention?
 - 7 Explain following commands: a) date b) ls c) rmdir
 - 8 Explain all control structures in details.

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